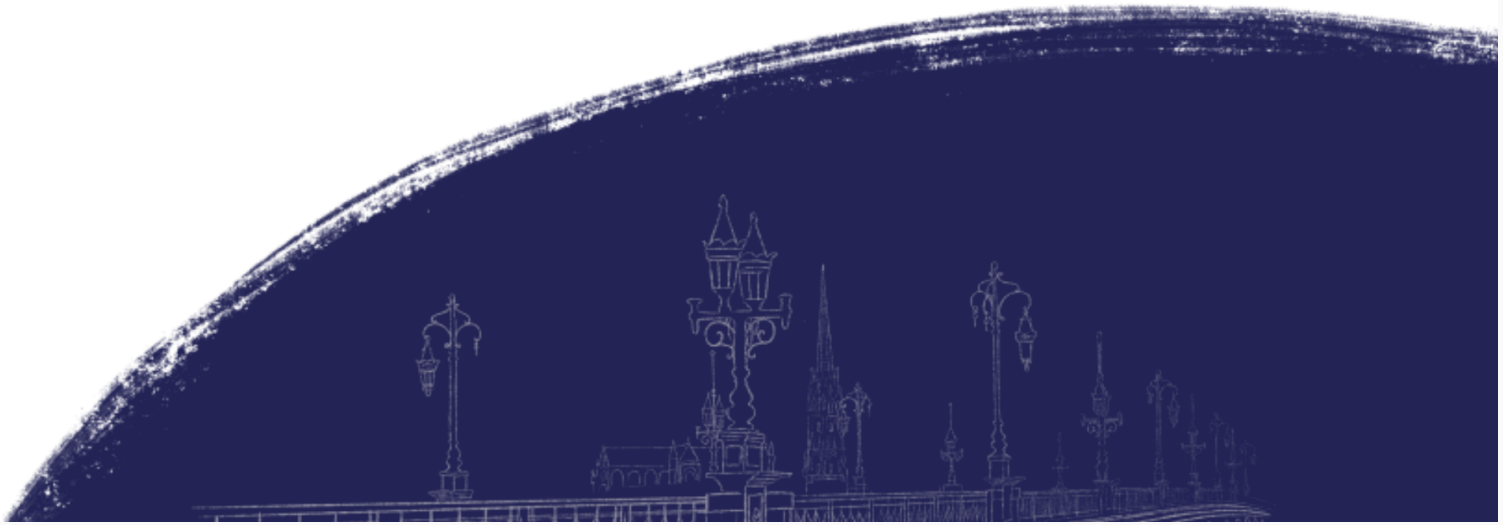
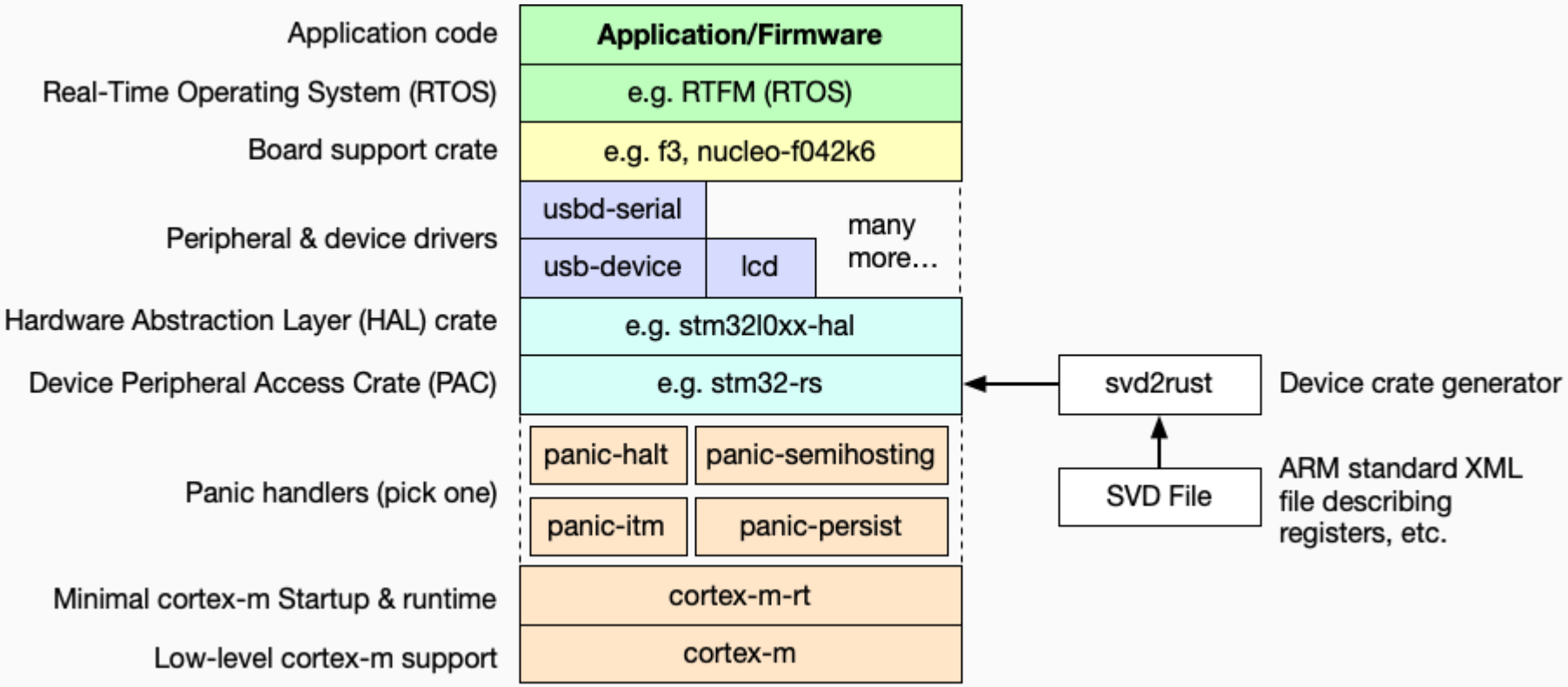
Rust pour le développement d'applications métier haut-niveau ! 🦀



Rust 🦀 est connu pour être un langage qui favoriserait le "bas-niveau"...



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Pour des masochistes férus d'optimisations mémoire...

À la syntaxe souvent qualifiée d'exigeante, voire de "cryptique"...

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Coercing Arc<Mutex<Option<Box<MyStruct>>>>> to Arc<Mutex<Option<Box<dyn Trait>>>>> won't work

Asked 4 years, 4 months ago Modified 4 years, 4 months ago Viewed 1k times

3

I'm trying to store a dyn trait inside `Arc<Mutex<Option<Box<>>>>>`, however for some reason it won't work

rust Copy

```rust
use std::sync::{Arc, Mutex};

trait A{}

struct B{}

impl A for B{}

struct H{
```

Th

Fe

BDX I/O

HEY, J'SUIS PAS VENUE ICI POUR
SOUFFRIR, OK ???

Tout ceci est p'tet vrai, mais que c'est pas pour ça que Rust m'intéresse

🦀Rust🦀
pour des applications métiers

BDX I/O

Un système de types expressif

La base: Structs, et "Newtypes"

Un cas ultra-classique:

```rust
pub struct User {
    pub email: String,
}

fn ma_fonction(user: User) {
  if !user.email.contains("@") {
     // Gestion de l'erreur, etc.
  }
  // On continue...

  // Possible, mais pas voulu -> Bug !
  let city: String = user.email
}
```

Encoder la logique métier, le rêve ! 😌

```rust
use crate::domain::UserEmail;

pub struct User {
    // Attendez…c'est quoi ça ?
    pub email: UserEmail,
}
```

On "impl"émente la logique pour traiter les cas :

```rust
use validator::ValidateEmail;

#[derive(Debug)]
pub struct UserEmail(String);

impl UserEmail {
    pub fn parse(s: String) -> Result<UserEmail, String> {
        if s.validate_email() {
            Ok(Self(s))
        } else {
            Err(format!("{} is not a valid user email", s))
        }
    }
}
```

Comment on valide ça ? Avec des tests ! 🤩

```rust
#[cfg(test)]
mod tests {
    #[test]
    fn email_missing_at_symbol_is_rejected() {
        let email = "stephanedomain.com".to_string();
        assert_err!(UserEmail::parse(email));
    }

    #[test]
    fn email_missing_subject_is_rejected() {
        let email = "@domain.com".to_string();
        assert_err!(UserEmail::parse(email));
    }
}
```

Un newtype n'est pas égal à un autre newtype !

```rust
pub struct Address(String);
pub struct UserEmail(String);

pub struct User { pub email: UserEmail }

let my_user = User {
  // 💥Erreur ! Pas un UserEmail, mais un String !
  email: "steph@mydomain.com"
}

let my_user = User {
  // 💥Erreur ! Pas un UserEmail, mais un Address !
  email: Address("steph@mydomain.com")
}
```

Il faut utiliser le newtype prévu :

```rust
use crate::domain::UserEmail;

pub struct User {
    pub email: UserEmail,
}

let my_user = User {
  // Ok 👍
  email: UserEmail::parse("steph@mydomain.com")?;
}
// Note: Pour l'instant le `?` c'est "magique" 🪄
```

ALEXIS KING

Parse, don't validate

2019-11-05 • functional programming, haskell, types

Historically, I've struggled to find a concise, simple way to explain what it means to practice type-driven design. Too often, when someone asks me "How did you come up with this approach?" I find I can't give them a satisfying answer. I know it didn't just come to me in a vision—I have an iterative design process that doesn't require plucking the "right" approach out of thin air—yet I haven't been very successful in communicating that process to others.

However, about a month ago, I was reflecting on Twitter about the differences I experienced parsing JSON in statically- and dynamically-typed languages, and finally, I realized what I was looking for. Now I have a single, snappy slogan that encapsulates what type-driven design means to me, and better yet, it's only three words long:

Parse, don't validate.

https://lexi-lambda.github.io/blog/2019/11/05/parse-don-t-validate/

Functional State Machines in Rust: Typestate and Newtype Patterns (Experience Report)

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Abstract

We investigate the effects on code quality of the design patterns typestate and newtype in the language Rust, by conducting three case studies from production code, evaluated through expert interviews, static code analysis and benchmarks. The findings indicate that: (1) Typestate improves code faultlessness and testability, but comes at the cost of more boilerplate code and can degrade readability. Code with extensive branching logic and many invariants likely benefits most from the pattern. (2) Newtype combined with the "Parse, don't validate" principle improves code quality at a low cost and prevents invalid states at runtime.

CCS Concepts: • **Software and its engineering** → *Patterns.*

Keywords: Rust, design patterns, typestate, newtype, functional core imperative shell, code quality, experience report

ACM Reference Format:

Most catalogued design patterns as of now originate in OOP, for example the Gang of Four (GoF) patterns [6]. Many GoF patterns become obsolete or unimplementable when transferred to Rust, since it is more expressive and provides native constructs for the problems those patterns solve.

Rust-specific patterns such as typestate and newtype have emerged. The Rust open source community documents a few of them in rust-unofficial/patterns. Crichton proposes four recurring functional design patterns in Rust and demonstrates their benefits through constructed case studies [4]. These resources provide useful practical guidance but lack evidence on the patterns' impact on code quality.

5 Conclusion

We conducted a case study on three use cases from Otto's backend applications, guided by a method built on SQuaRE criteria and combining execution-time benchmarks, static code analysis and expert interviews.

Neither typestate nor newtype incurred additional runtime cost. Typestate clearly improved testability and faultlessness by shifting error detection to compile time. However, its effect on readability is contested and is not captured by the static metrics. The additional boilerplate is perceived as cognitive overhead by some experts, while others find the explicit states easier to follow. Combined with the principle "Parse, don't validate", the newtype pattern introduced minimal downsides and code changes while improving almost all quality dimensions, making it broadly applicable.

Beyond addressing the limitations from subsection 4.4, several research directions open up. Because typestate's main downside is boilerplate, future studies should investigate macro rules or libraries that generate the state scaffolding and let developers focus on the logic of each state transition, similar to Duarte and Ravara [5]. Future work could furthermore strengthen the link between business requirements and source code. Since encoding invariants already transfers

Impureim sandwich by Mark Seemann

Pronounced 'impurium sandwich'.

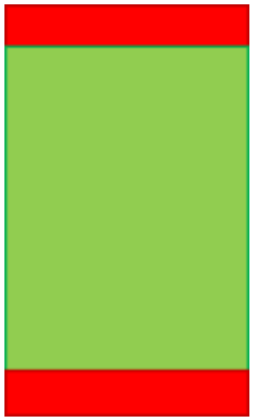
Since January 2017 I've been singing the praise of the *impure/pure/impure* sandwich, but I've never published an article that defines the term. I intend this article to remedy the situation.

Functional architecture

In a functional architecture, pure functions can't call impure actions. On the other hand, as Simon Peyton Jones observed in a lecture, *observing the result of pure computation is a side-effect*. In practical terms, *executing* a pure function is also impure, because it happens non-deterministically. Thus, even for a piece of software written in a functional style, the entry point must be impure.

While pure functions can't call impure actions, there's no rule to prevent the obverse. Impure actions *can* call pure functions.

Therefore, the best we can ever hope to achieve is an impure entry point that calls pure code and impurely reports the result from the pure function.



The flow of code here goes from top to bottom:

1. Gather data from impure sources.
2. Call a pure function with that data.
3. Change state (including user interface) based on return value from pure function.

This is the *impure/pure/impure* sandwich.

L'A et l'Ω du Type Driven Dev...

Le Pattern Matching

BDX I/O

Exemple avec un enum tout bête :

```
pub enum LoginError {
    AuthenticationError(String),
    SSOErrorCode(u32),
    UnexpectedError(String),
}
```

L'opérateur `match`

```
fn my_function(e: LoginError) {
    match e {}
          +-------------------------------------------------------------------------------+
}         |missing match arm: `AuthenticationError(_)`, `SSOErrorCode(_)` and             |
          |`UnexpectedError(_)` not covered (rust-analyzer E0004)                         |
          |-------------------------------------------------------------------------------|
          |non-exhaustive patterns: `LoginError::AuthenticationError(_)`, `LoginError::   |
          |SSOErrorCode(_)` and `LoginError::UnexpectedError(_)` not covered              |
          |the matched value is of type `LoginError`                                      |
          |                                                                               |
          |Related information:                                                           |
          |                                                                               |
          |  * main.rs#463,10: `LoginError` defined here                                  |
          |  * main.rs#470,12: ensure that all possible cases are being handled by       |
          |adding a match arm with a wildcard pattern, a match arm with multiple or-      |
          |patterns as shown, or multiple match arms: ` {                                 |
          |        LoginError::AuthenticationError(_) | LoginError::SSOErrorCode(_) |     |
          |LoginError::UnexpectedError(_) => todo!(),                                     |
          |    }`                                                                         |
          |                                                                               |
          | (rustc E0004)                                                                 |
          |-------------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                               |
          +-------------------------------------------------------------------------------+
```

BDX I/O

Fort heureusement on a ce qu'il faut avec le LSP de Rust 😌

```
fn my_function(e: LoginError) {
    match e {}
}           +Choose action------+
            |1. Fill match arms |
            +-------------------+
```

Et y a plus qu'à remplir !

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(_) => todo!(),
        LoginError::SSOErrorCode(_) => todo!(),
        LoginError::UnexpectedError(_) => todo!(),
    }
}
```

On peut laisser la vraie implem' pour plus tard

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(_) => todo!("Ah, pas cool 😱"),
        LoginError::SSOErrorCode(_) => todo!("Oulà ! Encore moins cool 😓"),
        LoginError::UnexpectedError(_) => todo!("Alors là jpp, moi… 😅"),
    }
}
```

On peut être arbitrairement complexe dans le matching

```
fn my_function(e: LoginError) {
    match e {
        LoginError::AuthenticationError(auth)
          if auth.contains("network") => todo!("Ah, pas cool 😱")
        LoginError::SSOErrorCode(c)
          if c > 42 && c < 1337 => todo!("Oulà ! Encore moins cool 😓"),
        LoginError::UnexpectedError(error)
          if error.is_empty() => todo!("Alors là jpp, moi… 😅"),
    }
}
```

Mais le compilateur veille...

```
fn my_function(e: LoginError) {
    match e {
          +-------------------------------------------------------------------------------+
        Lo|missing match arm: `AuthenticationError(_)` not covered (rust-analyzer E0004) |
          |-------------------------------------------------------------------------------|
        } |non-exhaustive patterns: `LoginError::AuthenticationError(_)` not covered      |
        Lo|the matched value is of type `LoginError`                                      |
        Lo|                                                                               |
    }     |Related information:                                                           |
}         |                                                                               |
          |  * main.rs#463,10: `LoginError` defined here                                  |
          |  * main.rs#475,76: ensure that all possible cases are being handled by       |
          |adding a match arm with a wildcard pattern or an explicit pattern as shown: `,|
          |        LoginError::AuthenticationError(_) => todo!()`                         |
          |                                                                               |
          | (rustc E0004)                                                                 |
          |-------------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                               |
          +-------------------------------------------------------------------------------+
```

Voyez-vous le problème, ici ?

```rust
pub enum Direction {
    North,
    South,
    East,
    West,
}
type Coordinates<'a> = (&'a u32, &'a u32);

pub fn move_player((&x, &y): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
        ((0, 0), _) => todo!("Origine du plan, rien à faire"),
        ((13, 10), Direction::South) => todo!("Ah non y a l'océan par-là !"),
    }
}
```

Le compilateur est plus vigilant que vous 😸

```
pub fn move_player((&x: u32, &y: u32): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
          +-----------------------------------------------------------------------------+
        (((|non-exhaustive patterns: `((1_u32..=12_u32, _), _)` and `((14_u32..=u32::MAX,|
        (((|_), _)` not covered                                                         |
    }     |the matched value is of type `((u32, u32), &Direction)`                     |
}         |                                                                             |
          |Related information:                                                         |
          |                                                                             |
          |  * main.rs#412,77: ensure that all possible cases are being handled by      |
          |adding a match arm with a wildcard pattern, a match arm with multiple or-    |
          |patterns as shown, or multiple match arms: `,                                |
          |        ((1_u32..=12_u32, _), _) | ((14_u32..=u32::MAX, _), _) => todo!()`   |
          |                                                                             |
          | (rustc E0004)                                                               |
          |-----------------------------------------------------------------------------|
          |https://doc.rust-lang.org/error-index.html#E0004                             |
          +-----------------------------------------------------------------------------+
```

BDX I/O

Et finalement, tout s'arrange 😉

```rust
pub enum Direction {
    North,
    South,
    East,
    West,
}
type Coordinates<'a> = (&'a u32, &'a u32);

pub fn move_player((&x, &y): Coordinates, direction: &Direction) {
    match ((x, y), direction) {
        ((0, 0), _) => todo!("Origine du plan, rien à faire"),
        ((13, 10), Direction::South) => todo!("Ah non y a l'océan par-là !"),
        ((_,_) , _) => todo!("Autres cas à traiter plus tard")
    }
}
```

Les "Boîtes" de Rust

BDX I/O

Moment Kahoot !

BDX I/O

Hé oui, c'est bien Null:

Speaking at a software conference in 2009, Tony Hoare hyperbolically apologized for inventing the null reference: [26][27]

> I call it my billion-dollar mistake. It was the invention of the null reference in 1965. At that time, I was designing the first comprehensive type system for references in an object oriented language (ALGOL W). My goal was to ensure that all use of references should be absolutely safe, with checking performed automatically by the compiler. But I couldn't resist the temptation to put in a null reference, simply because it was so easy to implement. This has led to innumerable errors, vulnerabilities, and system crashes, which have probably caused a billion dollars of pain and damage in the last forty years.[28]

En Javascript (et autres...):

En Rust, plus de doute !

```
// get_user_by_email:
//
//   UserEmail -> Option<User>
//
match get_user_by_email(user_email).await {

    Some(user) => todo!("rajouter le cas où l'utilisateur existe"),

    None => todo!("créer le user ? retourner une erreur ? vous décidez !"),

}
```

Moment Kahoot 2 (Le Retour !)

BDX I/O

Why Do Some Developers Consider `Try-Catch` to Be Problematic?

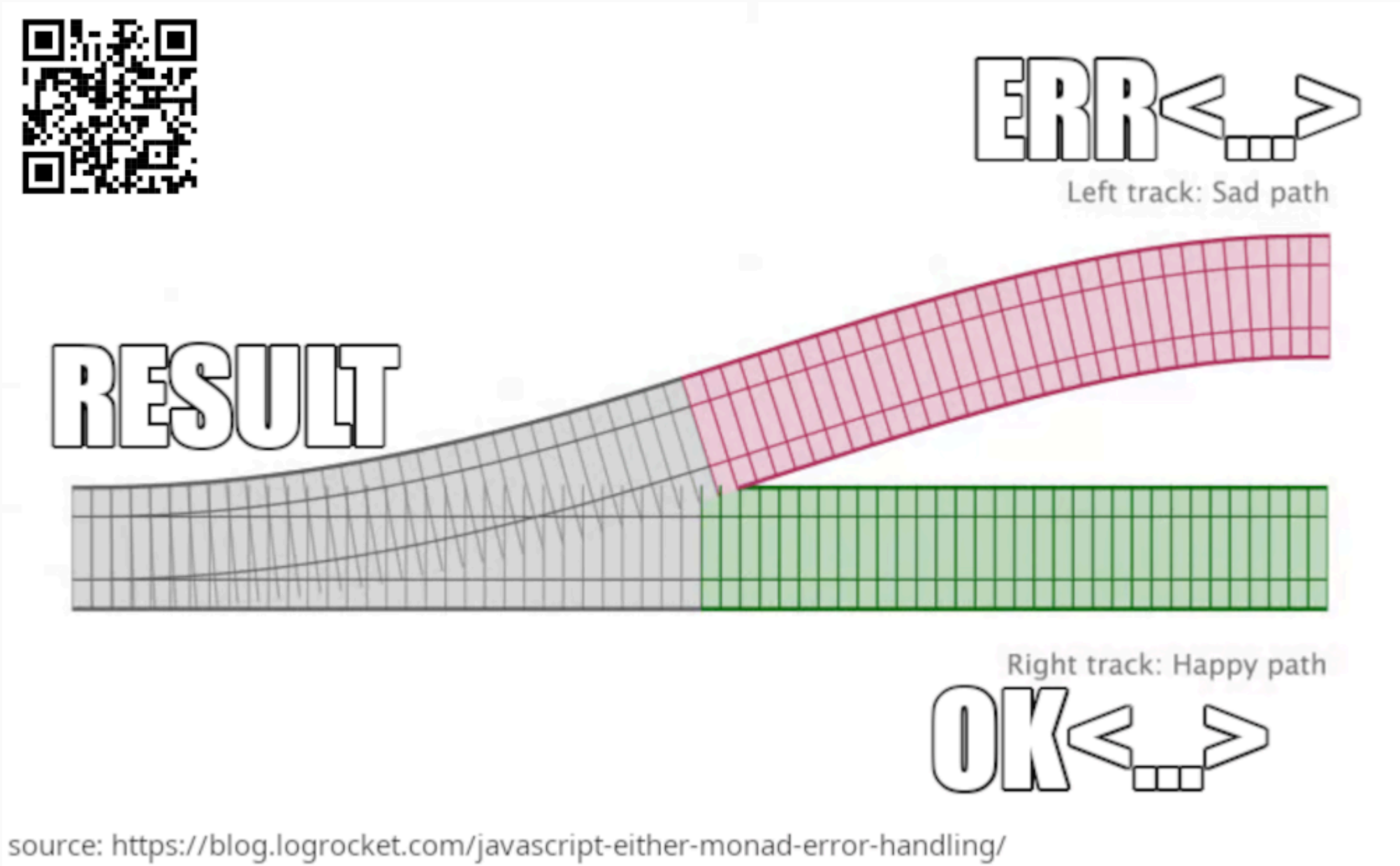
The sentiment that `try-catch` is "evil" stems from several potential pitfalls:

- **Masking Errors**: Catching exceptions too broadly can hide underlying problems, making debugging difficult.
- **Performance Overhead**: Exception handling can be computationally expensive, especially if exceptions are frequently thrown and caught.
- **Code Complexity**: Excessive `try-catch` blocks can clutter code and make it harder to read and maintain.
- **Control Flow**: Using exceptions for normal control flow is generally bad practice and can lead to unexpected behavior.

https://deepdatawithmivaa.com/why-is-try-catch-evil

Certains langages font ce qu'ils peuvent… 😅

D'autres, comme Rust, plongent dans le multivers ! 💪



source: https://blog.logrocket.com/javascript-either-monad-error-handling/

Rappelez-vous `UserEmail` ! 🤓

```rust
use validator::ValidateEmail;

#[derive(Debug)]
pub struct UserEmail(String);

impl UserEmail {                +-------------------------+
    pub fn parse(s: String) -> |Result<UserEmail, String>| {
                                +-------------------------+
        if s.validate_email() {
           +-----------+
           |Ok(Self(s))|
           +-----------+
        } else {
           +-----------------------------------------------+
           |Err(format!("{} is not a valid user email", s))|
           +-----------------------------------------------+
        }
    }
}
```

De plus, Un Result étant un Enum, on peut le `match` 🤩

```
// validate_credentials:
//
//   Credentials -> Result<Uuid, LoginError>
//
match validate_credentials(credentials).await {
}                       +Choose action -----+
                        |1. Fill match arms |
                        +-------------------+
```

Et y a plus qu'à remplir, ou à `todo` ! 🤷

```
// validate_credentials:
//
//   Credentials -> Result<Uuid, LoginError>
//
match validate_credentials(credentials).await {

    Ok(_) => todo!("rajouter le cas où tout est ok"),

    Err(_) => todo!("rajouter la gestion d'erreur"),

}
```

Finalement, on met le code final:

```
match validate_credentials(credentials).await {
  Ok(user_id) => {
    return HttpResponse::SeeOther()
      .insert_header((LOCATION, "/admin/dashboard"))
      .finish());
  }
  Err(e) => {
    return login_redirect(LoginError::AuthError(e));
  }
}
```

"On est obligés de `match` à chaque fois un Result ?" 🤔

Non, on peut utiliser (entre autres) `.unwrap()`, pour ça:

```rust
// Dans ce cas-ci, on est persuadés qu'on
// aura toujours un port disponible
let port = listener.local_addr().unwrap().port();
```

Mais pourquoi on s'inflige `match`, alors ? ☝️🤓

...et même les plus gros se font avoir 💥

When the bad file with more than 200 features was propagated to our servers, this limit was hit — resulting in the system panicking. The FL2 Rust code that makes the check and was the source of the unhandled error is shown below:

```rust
71      /// Fetch edge features based on `input` struct into [`Features`] buffer.
72      pub fn fetch_features(
73          &mut self,
74          input: &dyn BotsInput,
75          features: &mut Features,
76      ) -> Result<(), (ErrorFlags, i32)> {
77          // update features checksum (lower 32 bits) and copy edge feature names
78          features.checksum &= 0xFFFF_FFFF_0000_0000;
79          features.checksum |= u64::from(self.config.checksum);
80          let (feature_values, _) = features
81              .append_with_names(&self.config.feature_names)
82              .unwrap();
```

This resulted in the following panic which in turn resulted in a 5xx error:

```
thread fl2_worker_thread panicked: called Result::unwrap() on an Err value
```

https://blog.cloudflare.com/18-november-2025-outage/

Réservez `.unwrap()` pour le code de test et apprenez à propager ou gérer correctement les `Result`

```
fn my_function() -> Result<PortNumber, Error> {
  /* In this case, we SHOULD have an IP address. */
  /* If not, bubble up the error upwards */
  let port = listener.local_addr()?.port();
  return port;
}
```

Rappel que Clippy est votre ami

unwrap_used

restriction allow –

What it does

Checks for `.unwrap()` or `.unwrap_err()` calls on `Results` and `.unwrap()` call on `Options`.

Why restrict this?

It is better to handle the `None` or `Err` case, or at least call `.expect(_)` with a more helpful message. Still, for a lot of quick-and-dirty code, `unwrap` is a good choice, which is why this lint is `Allow` by default.

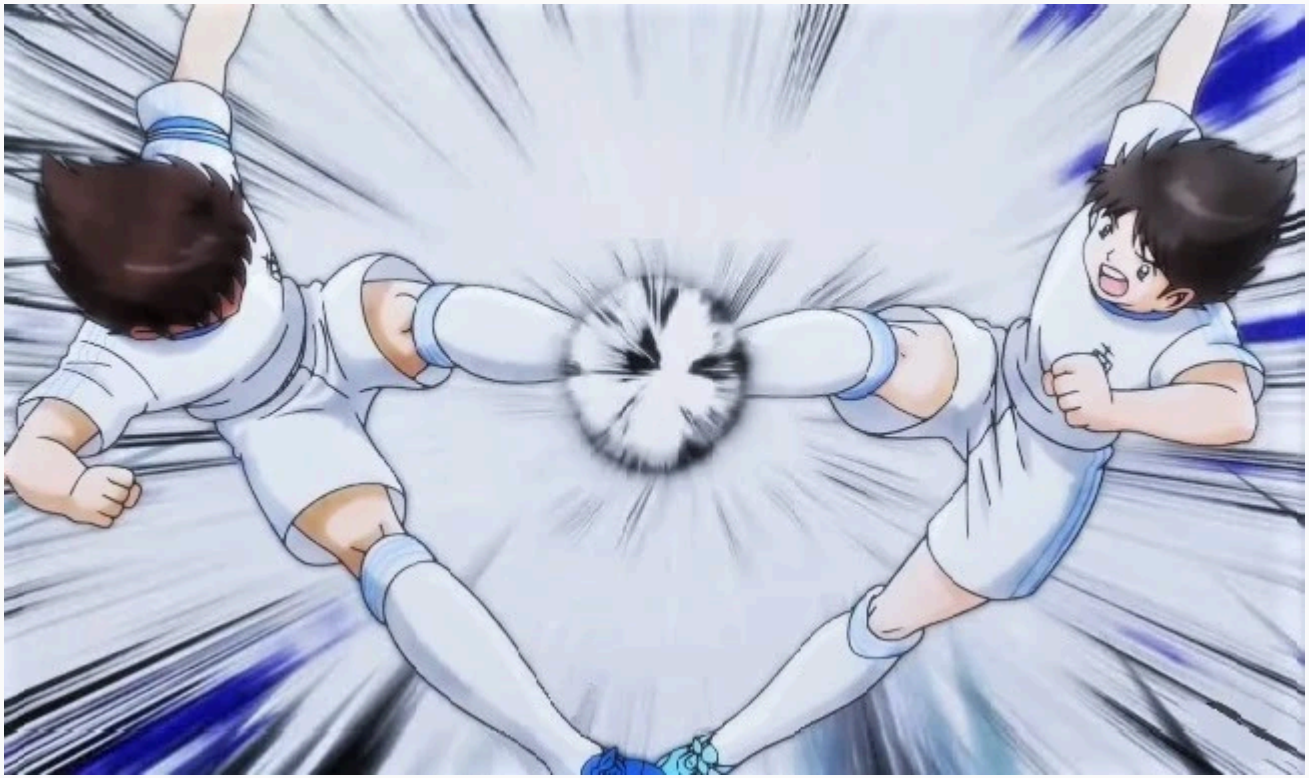
`result.unwrap()` will let the thread panic on `Err` values. Normally, you want to implement more sophisticated error handling, and propagate errors upwards with `?` operator.

Even if you want to panic on errors, not all `Errors` implement good messages on display. Therefore, it may be beneficial to look at the places where they may get displayed. Activate this lint to do just that.

D'ailleurs, en parlant d'erreurs...

Les jumeaux surdoués de la gestion d'erreur

anyhow et thiserror



BDX I/O

anyhow, les erreurs faciles à créer :

```rust
use anyhow::{Context, Result};

/*  Le Result ne précise même plus le type d'erreur */
/*  car ce sera un `anyhow::Error` 👌 */
fn main() -> Result<()> {

  /*  .context permet d'emballer l'erreur avec…du contexte 😁 */
  it.detach().context("Failed to detach the thing")?;

  /*  L'utilisation de `?` veut simplement dire: */
  /*  Si c'est une `Err` alors on bubble up ! */
  /*  Sinon on "ouvre" la boîte `Ok` et on prend son contenu */
  let content = std::fs::read(path)?;
}
```

`thiserror`, les erreurs faciles à décrire :

```rust
use thiserror::Error;
#[derive(Error, Debug)]
pub enum DataStoreError {
  #[error("data store disconnected")]
  Disconnect(#[from] io::Error),
  #[error("the data for key `{0}` is not available")]
  Redaction(String),
  #[error("invalid ({expected:?}, found {found:?})")]
  InvalidHeader {
      expected: String,
      found: String,
  },
  #[error("unknown data store error")]
  Unknown
}
```

`thiserror` pour tous les types ? `derive_more` !

```rust
use derive_more::{Add, Display, From, Into};

#[derive(PartialEq, From, Add)]
struct MyInt(i32);

#[derive(PartialEq, From, Into)]
struct Point2D {
    x: i32,
    y: i32,
}

#[derive(PartialEq, From, Add, Display)]
enum MyEnum {
    #[display("int: {_0}")]
    Int(i32),
    Uint(u32),
    #[display("nothing")]
    Nothing,
}
```

Vous trouvez ça lourd de devoir déclarer un enum ? eros à la rescousse !

```rust
use eros::IntoUnion;
use std::{io, num::ParseIntError};

fn load_port(path: &str) -> eros::Result<u16, (io::Error, ParseIntError)> {
    let contents = std::fs::read_to_string(path).union()?;
    contents.trim().parse().union()
}

fn port_or_default(path: &str) -> eros::Result<u16, (ParseIntError,)> {
    load_port(path).recover::<io::Error, _>(|error| {
        eprintln!("{error:?}; using port 8080");
        8080
    })
}
```

Traits, la POO turbo-chargée

On définit ce qu'est un PaymentProcessor

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}
```

On veut ensuite implémenter "notre" PaymentProcessor

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
```

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
```

```
   +-----------------------------------------------------------------------------+
   |not all trait items implemented, missing: `send_payment`                    |
   |missing `send_payment` in implementation                                     |
   |                                                                             |
   |Related information:                                                         |
   |                                                                             |
   |  * main.rs#422,5: `send_payment` from trait                                 |
   |  * main.rs#431,47: implement the missing item: `fn send_payment(_:          |
   |&RecipientEmail, _: &SenderEmail) -> std::result::Result<(), anyhow::Error> { |
   |todo!() }`: `fn send_payment(_: &RecipientEmail, _: &SenderEmail) ->         |
   |std::result::Result<(), anyhow::Error> { todo!() }`                          |
   +-----------------------------------------------------------------------------+
```

Comme pour les enums, on peut utiliser le puissant LSP de Rust

```
pub struct MyPaymentProcessor {}

impl PaymentProcessor for MyPaymentProcessor {}
      +Choose action --------------+
      |1. Implement missing members|
      +----------------------------+
```

Et on met ensuite l'implem' qu'on veut:

```rust
pub struct RecipientEmail(pub String);
pub struct SenderEmail(pub String);

pub trait PaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error>;
}

pub struct MyPaymentProcessor {}
impl PaymentProcessor for MyPaymentProcessor {
    fn send_payment(
        recipient: &RecipientEmail,
        sender: &SenderEmail,
    ) -> Result<(), anyhow::Error> {
        todo!("Payment instructions from {sender} to {recipient} go here !")
    }
}
```

Les traits "standards"

Un exemple simple: J'ai une `LoginError`, et je veux la déboguer

```
pub enum LoginError {
    AuthenticationError(String),
    UnexpectedError(String),
}
```

Une "fonction" super pratique: dbg !

```
let my_error = LoginError::AuthError("Oulà !");
dbg!(&my_error)
```

Sauf que LoginError n'est pas un type élémentaire...

```
+-------------------------------------------------------------+
|`LoginError` doesn't implement `std::fmt::Debug`             |
|add `#[derive(Debug)]` to `LoginError` or manually `impl     |
|std::fmt::Debug for LoginError`                              |
|                                                             |
|Related information:                                         |
|                                                             |
|  * error.rs#53,18: required by a bound in `Error`           |
|  * post.rs#18,1: consider annotating `LoginError` with      |
|    `#[derive(Debug)]`                                       |
|                                                             |
| (rustc E0277)                                               |
|-------------------------------------------------------------|
|https://doc.rust-lang.org/error-index.html#E0277             |
+-------------------------------------------------------------+
```

Dériver, simple et bien pratique dans 90% des cas :

```
#[derive(Debug)]
pub enum LoginError {
    AuthError(Error),
    UnexpectedError(Error),
}
```

Ok pour les cas de base, mais comment implémenter explicitement Debug ?

Encore une fois, le LSP est là pour nous aider 😍

```
pub enum LoginError {
    AuthError(String),
    UnexpectedError(String),
}

impl std::fmt::Debug for LoginError {}
              +Choose action --------------+
              |1. Implement missing members|
              +----------------------------+
```

On obtient une implémentation "par defaut", qu'on peut alors customiser 🤤

```rust
impl std::fmt::Debug for LoginError {
  fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    match self {
      Self::AuthError(arg0) =>
          f.debug_tuple("AuthError").field(arg0).finish(),
      Self::UnexpectedError(arg0) =>
          f.debug_tuple("UnexpectedError").field(arg0).finish(),
    }
  }
}
```

Et on y met ce qu'on veut (tant qu'on respecte le Trait)

```
impl std::fmt::Debug for LoginError {
    fn fmt(&self, f: &mut Formatter<'_>) -> Result {
        writeln!(f, "{}\n", e)?;
        let mut current = e.source();
        while let Some(cause) = current {
            writeln!(f, "Caused by:\n\t{}", cause)?;
            current = cause.source();
        }
        Ok(())
    }
}
```

Ce qui nous permet d'avoir un détail de l'erreur bien plus clair en debug :

```
Failed to log in user.
Caused by:
    Failed to retrieve user informations
Caused by:
    error returned from database
Caused by:
    table "user" does not exist
```

La "blanket implementation" d'un Trait pour un Type

Idée: je veux customiser la manière de loguer de mes variables 🤔

```rust
trait Loggable { fn log(&self); }

pub struct Cat { pub name: String, }

fn main() {
    let numbers = vec![1, 2, 3];
    numbers.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implémente Loggable pour u64 😫

    let s = String::from("Blanquette de Veau");
    s.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implémente Loggable pour String 😫

    let c = Cat { name: "Mimouchka".to_string(), };
    c.log(); // Erreur ! 💥
    // Ah, mais oui ! Faut que j'implémente Loggable pour Cat 😫
}
```

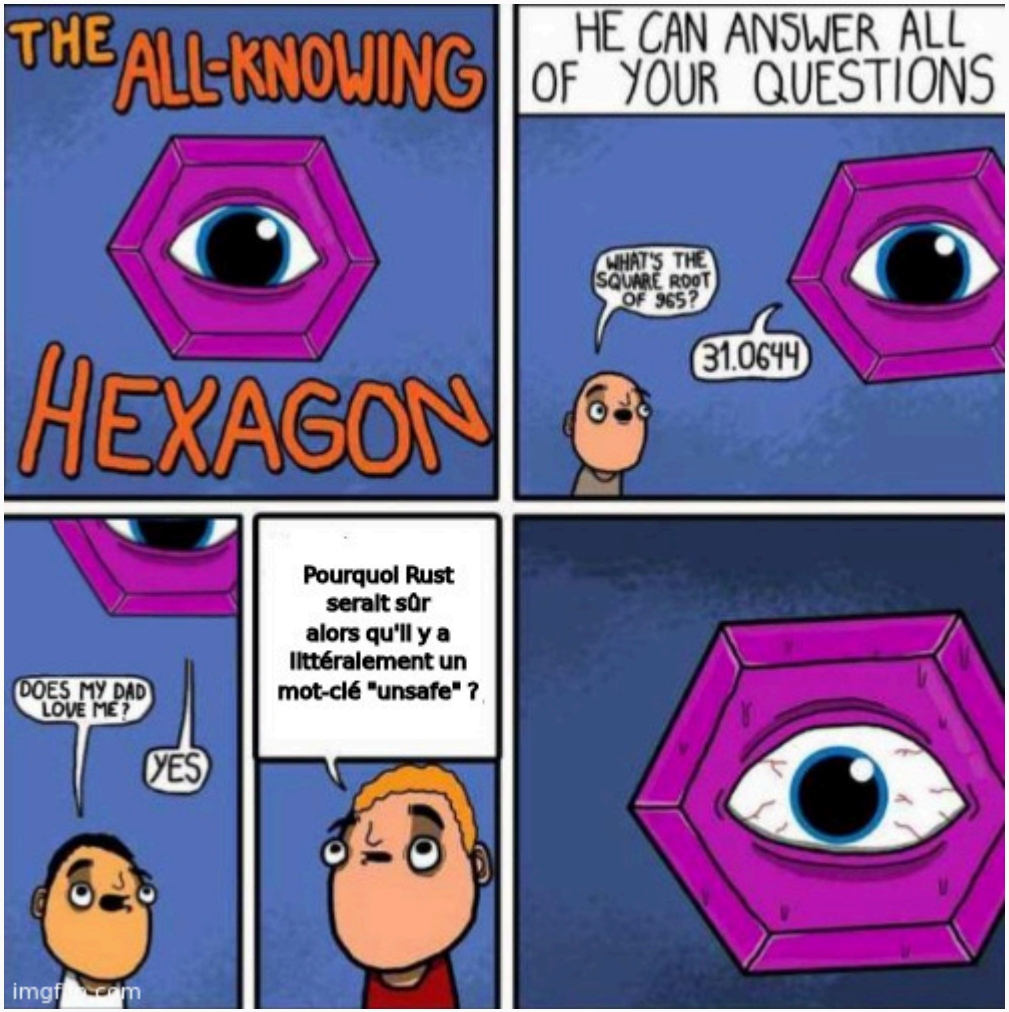
`impl<T>`, parce que nous le valons bien 💇

```rust
trait Loggable { fn log(&self); }

// Tout ce qui impl Debug va désormais automatiquement impl Loggable ! 🤩
impl<T> Loggable for T
where T: Debug {
    fn log(&self) { println!("[log] {:?}", self); }
}

#[derive(Debug)]
pub struct Cat { pub name: String, }

fn main() {
    let numbers = vec![1, 2, 3];
    numbers.log(); // Vec<i32> impl nativement Debug, et donc Loggable ! 😌

    let s = String::from("Blanket");
    s.log(); // String impl nativement Debug, et donc Loggable ! 😌

    let c = Cat { name: "Astra".to_string(), };
    c.log(); // Cat impl Debug (via derive), et donc Loggable ! 😌
}
```

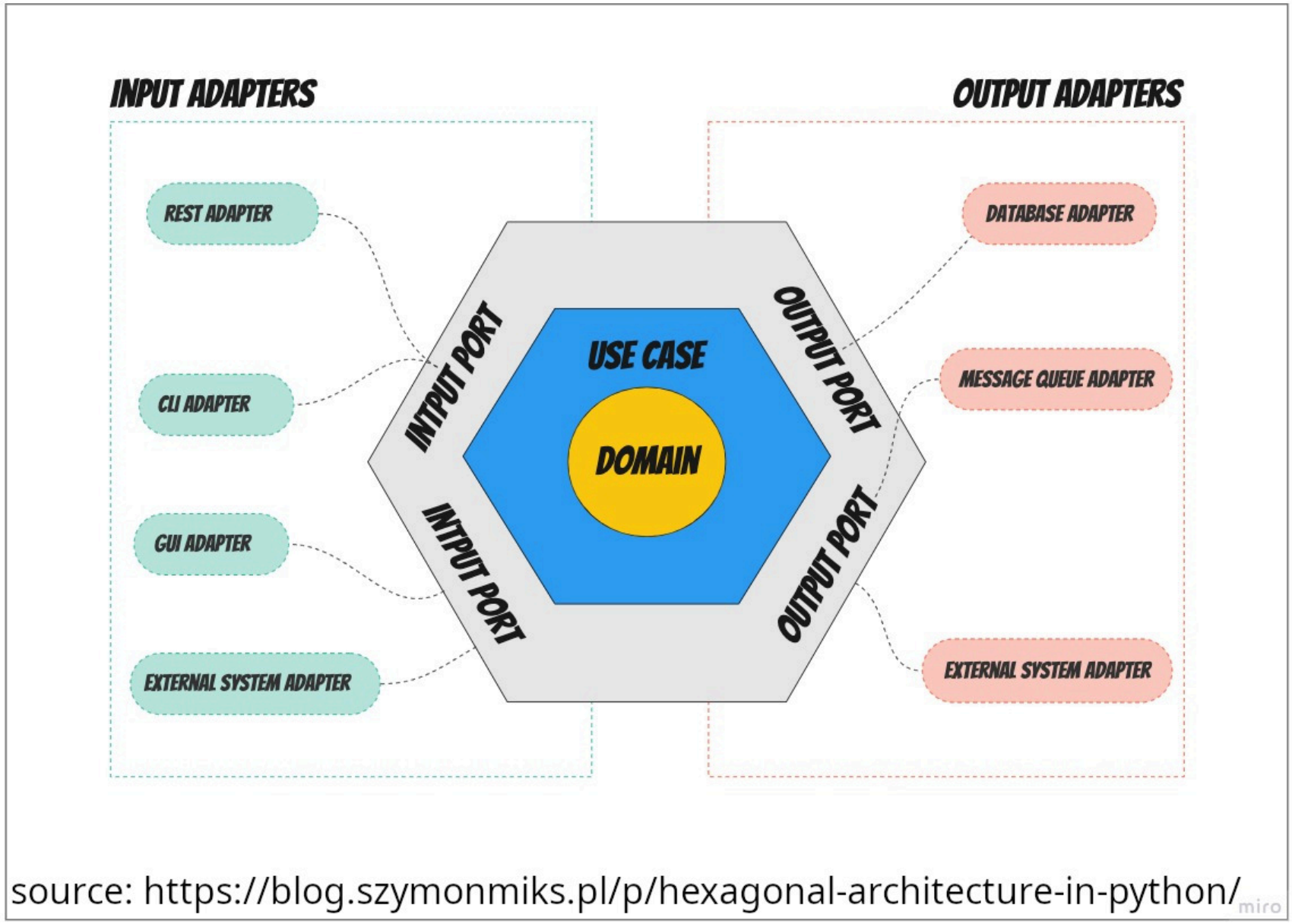
Et des types/traits/impl comme ça, vous en avez PLEIN, dans Rust 🤩

```
- `Option` (`Some<…>` ou `None`)
- `Debug` vs `Display`
- `Deref`
- `Drop`
- ...et plein d'autres (sans compter ceux de la communauté 🛡)
```

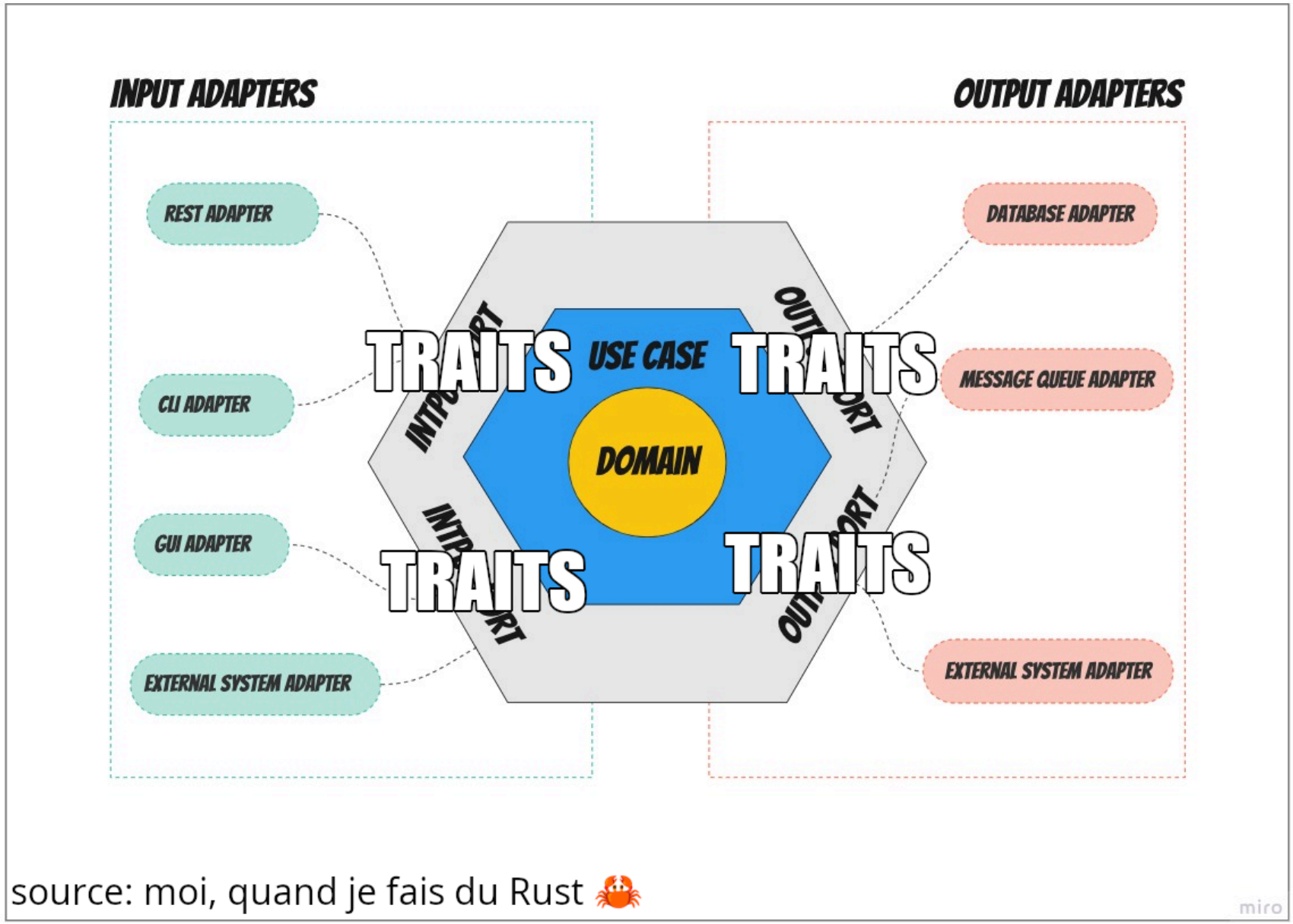
Architecture Hexagonale



Rappel à toutes fins utiles:



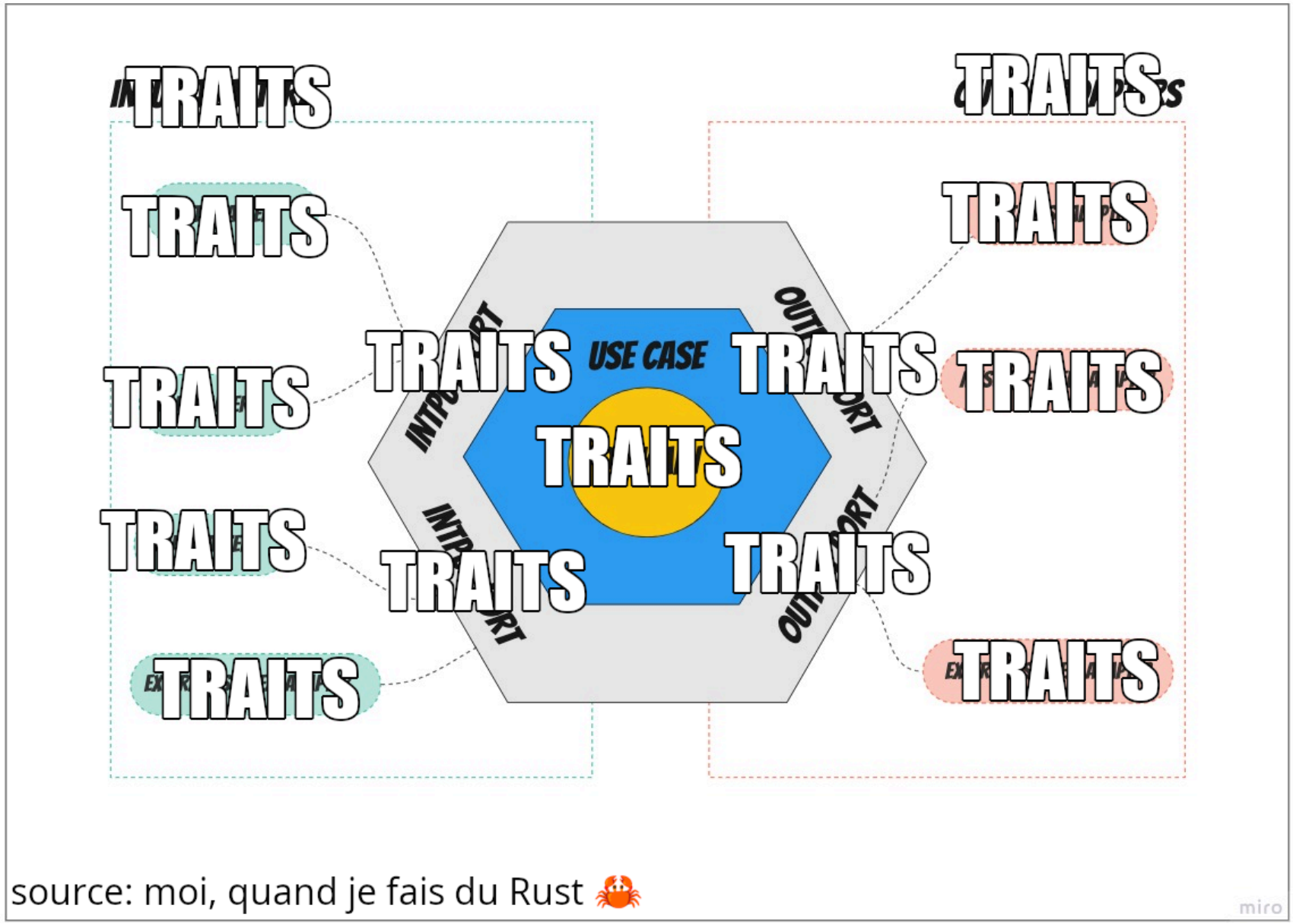
Donc, si vous avez bien suivi…



source: moi, quand je fais du Rust 🦀

BDX I/O

Mais, en vrai…


TRAITS
TRAITS
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USE CASE
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TRAITS

Et c'est ok !

```rust
fn process_data<F, B>(foo: F, bar: B) -> String
where
    F: Clone + Debug,
    B: Display + PartialEq,
{
    // Function implementation
    format!("{:?} - {}", foo, bar)
}
```

⬢ + 🦀 = 💓

```rust
pub async fn create_author<AR: AuthorRepository>(
    State(state): State<AppState<AR>>,
    Json(body): Json<CreateAuthorHttpRequestBody>,
) -> Result<ApiSuccess<CreateAuthorResponseData>, ApiError> {
    let domain_req = body.try_into_domain()?;
    state
        .author_repo
        .create_author(&domain_req)
        .await
        .map_err(ApiError::from)
        .map(|ref author| ApiSuccess::new(StatusCode::CREATED, author.into()))
}
```

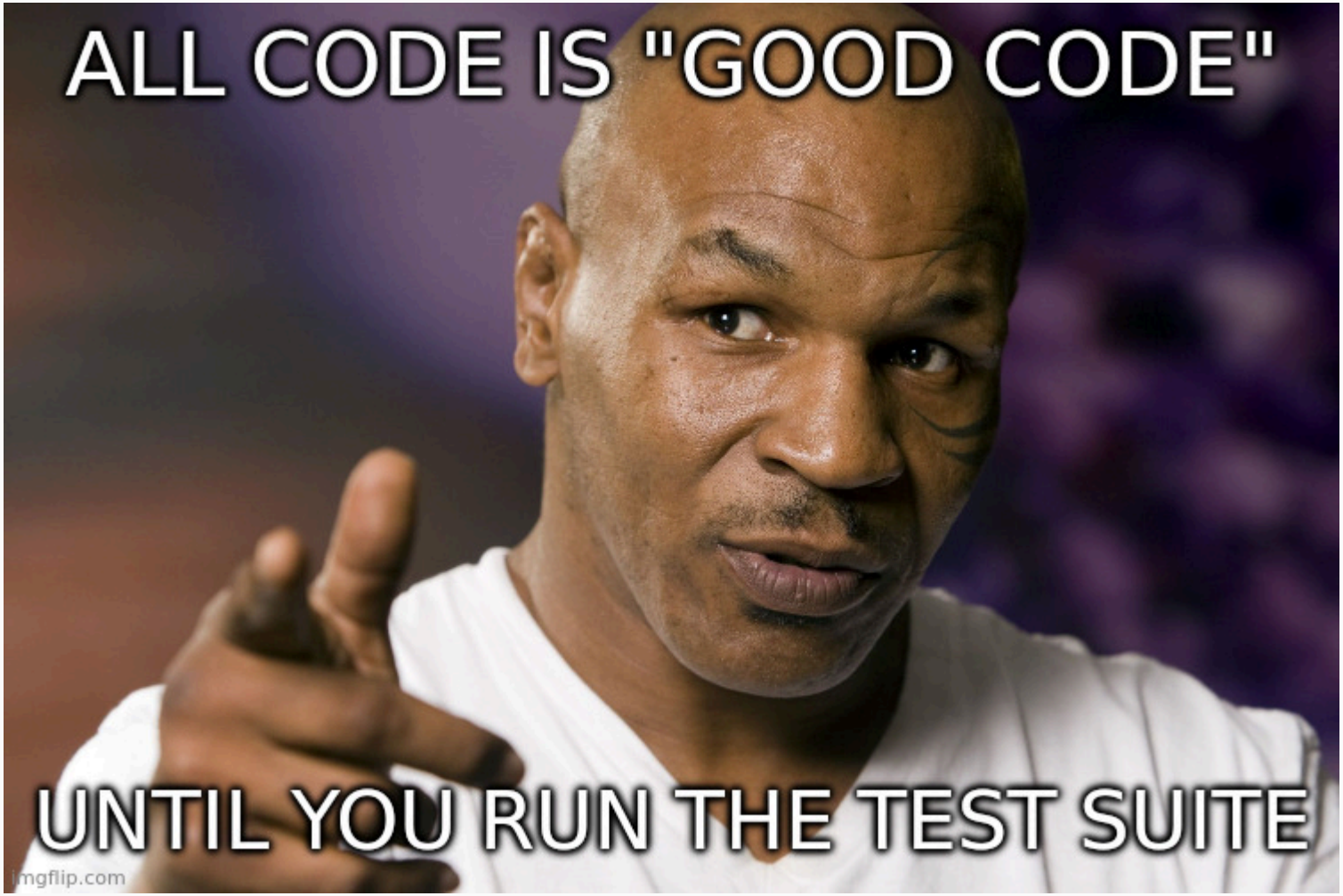
source: https://www.howtocodeit.com/guides/master-hexagonal-architecture-in-rust

Oh my.

Isn't it beautiful?

Doesn't your nervous system feel calmer to behold it?

Tester dans Rust



Doc et tests colocalisés avec le code 💕

```
…/temp-rust  trunk  v1.93.0  ?
(ins)› cargo test -q

running 2 tests
. 1/2
tests::test_adder_ko ── FAILED

failures:

─── tests::test_adder_ko stdout ───

thread 'tests::test_adder_ko' (186854) panicked at src/lib.rs:23:9:
assertion `left == right` failed
  left: 3
 right: 4
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace


failures:
    tests::test_adder_ko

test result: FAILED. 1 passed; 1 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.0
0s

error: test failed, to rerun pass `--lib`
```

```rust
/// Adder: Add two numbers.
/// ```
/// let x = 5;
/// let y = 5;
/// assert_eq!(mylib::adder(x,y), 10);
/// ```
pub fn adder<T>(x: T, y: T) -> T
where T: std::ops::Add<Output = T> {
    x + y
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn test_adder_ok() {
        assert_eq!(adder(1, 2), 3);
    }

    #[test]
    fn test_adder_ko() {
        assert_eq!(adder(1, 2), 4);
    }
}
```

BDX I/O

La doc des projets Rust est… *chef kiss* 👌

```rust
use mylib::adder;

fn main() {
println!("{}", adder(40, 2));
}
```

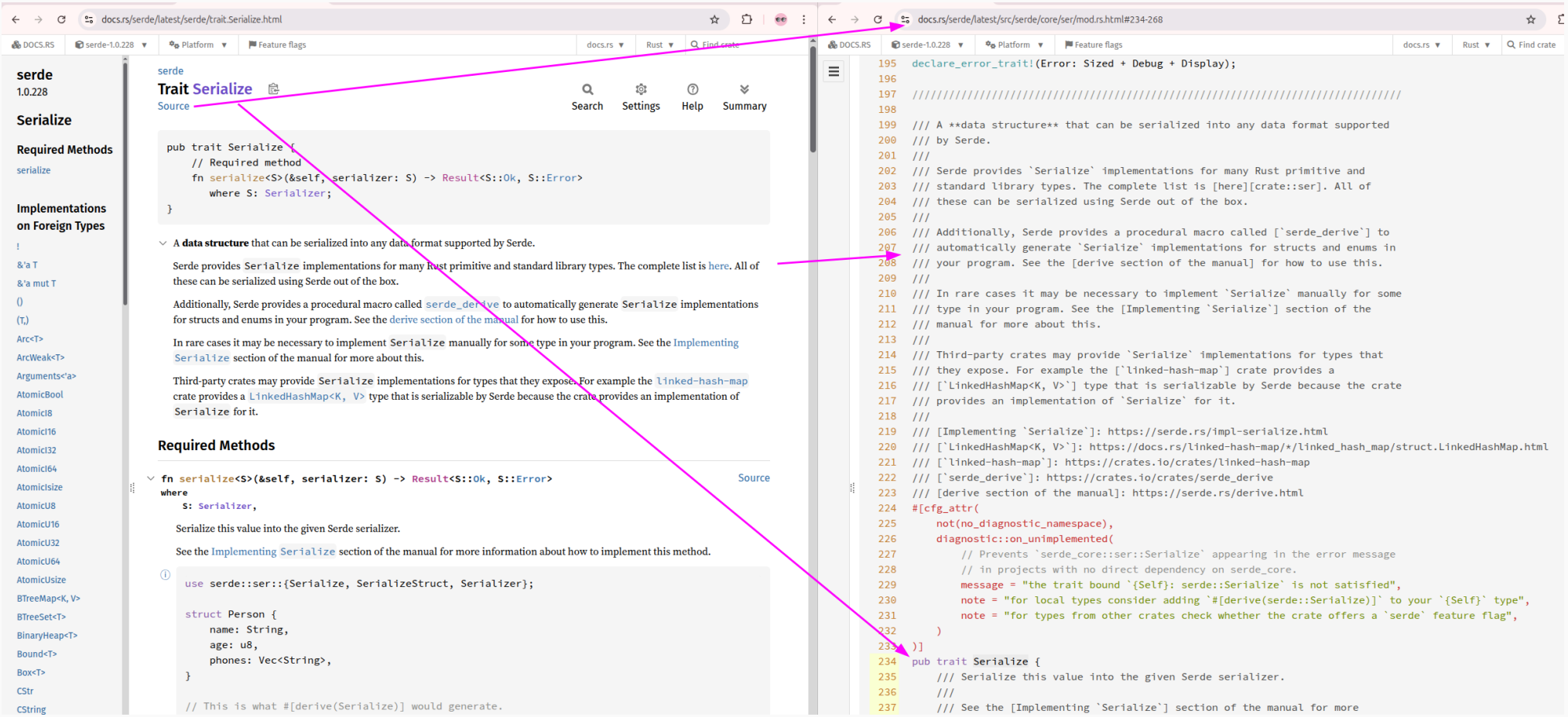
```rust
mylib

pub fn adder<T>(x: T, y: T) -> T
where
    T: std::ops::Add<Output = T>,
```

```
T = i32
```

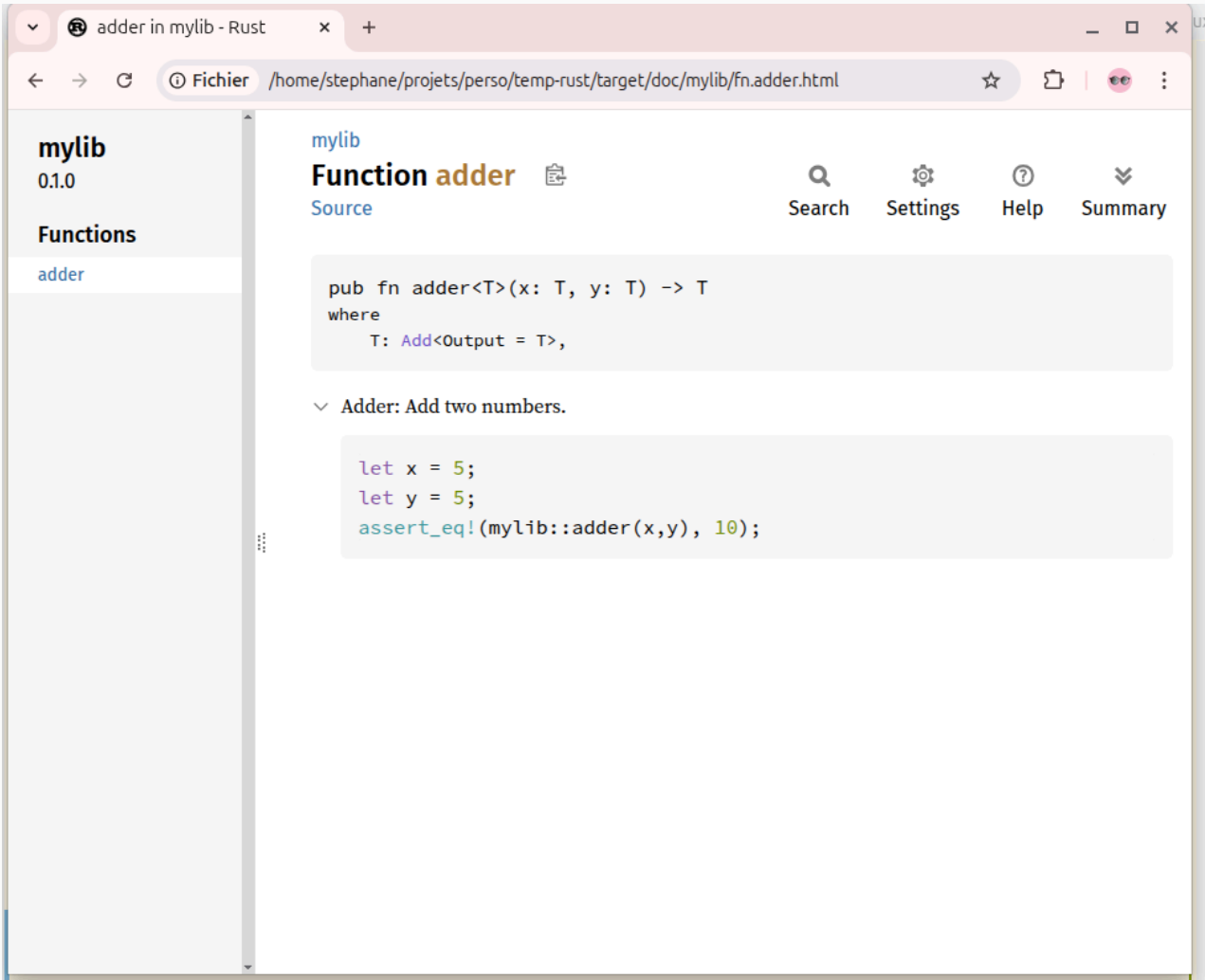
```rust
Adder: Add two numbers.
let x = 5;
let y = 5;
assert_eq!(mylib::adder(x,y), 10);
```

BDX I/O

TOUT est "as code":



cargo doc , tout simplement




```
…/temp-rust  trunk  v1.93.0 ?
(ins)› cargo doc -q

…/temp-rust  trunk  v1.93.0 ?
(ins)›
```

ux: client

```rust
/// Adder: Add two numbers.
///
/// ```
/// let x = 5;
/// let y = 5;
/// assert_eq!(mylib::adder(x,y), 10);
/// ```
pub fn adder<T>(x: T, y: T) -> T
where T: std::ops::Add<Output = T> {
    x + y
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn test_adder_ok() {
        assert_eq!(adder(1, 2), 3);
    }

    #[test]
    fn test_adder_ko() {
        assert_eq!(adder(1, 2), 4);
    }
}
```

BDX I/O

Oh, et ils sont *par défaut* testés par `cargo test` 😎

```
running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s


running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s


running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

all doctests ran in 0.15s; merged doctests compilation took 0.14s
```

```rust
/// Adder: Add two numbers.
///
/// let x = 5;
/// let y = 5;
/// assert_eq!(mylib::adder(x,y), 10);
///
pub fn adder<T>(x: T, y: T) -> T
where T: std::ops::Add<Output = T> {
    x + y
}
```

BDX I/O

Votre doc ne fait donc pas que décrire vos invariants, elle les contrôle ! 💪

```
running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s


running 1 test
src/lib.rs - adder (line 2) ... FAILED

failures:

---- src/lib.rs - adder (line 2) stdout ----
Test executable failed (exit status: 101).

stderr:

thread 'main' (196168) panicked at /tmp/rustdoctestQGu8L7/doctest_bundle_2024.rs:8:1:
assertion `left == right` failed
  left: 10
 right: 11
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace



failures:
    src/lib.rs - adder (line 2)

test result: FAILED. 0 passed; 1 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.0
0s
```

```rust
/// Adder: Add two numbers.
/// ```
/// let x = 5;
/// let y = 5;
/// assert_eq!(mylib::adder(x,y), 11);
/// ```
pub fn adder<T>(x: T, y: T) -> T
where T: std::ops::Add<Output = T> {
    x + y
}
```

BDX I/O

Tests d'Intégration ?

Facile ! On les met dans `/tests` (avec du code de setup si besoin)

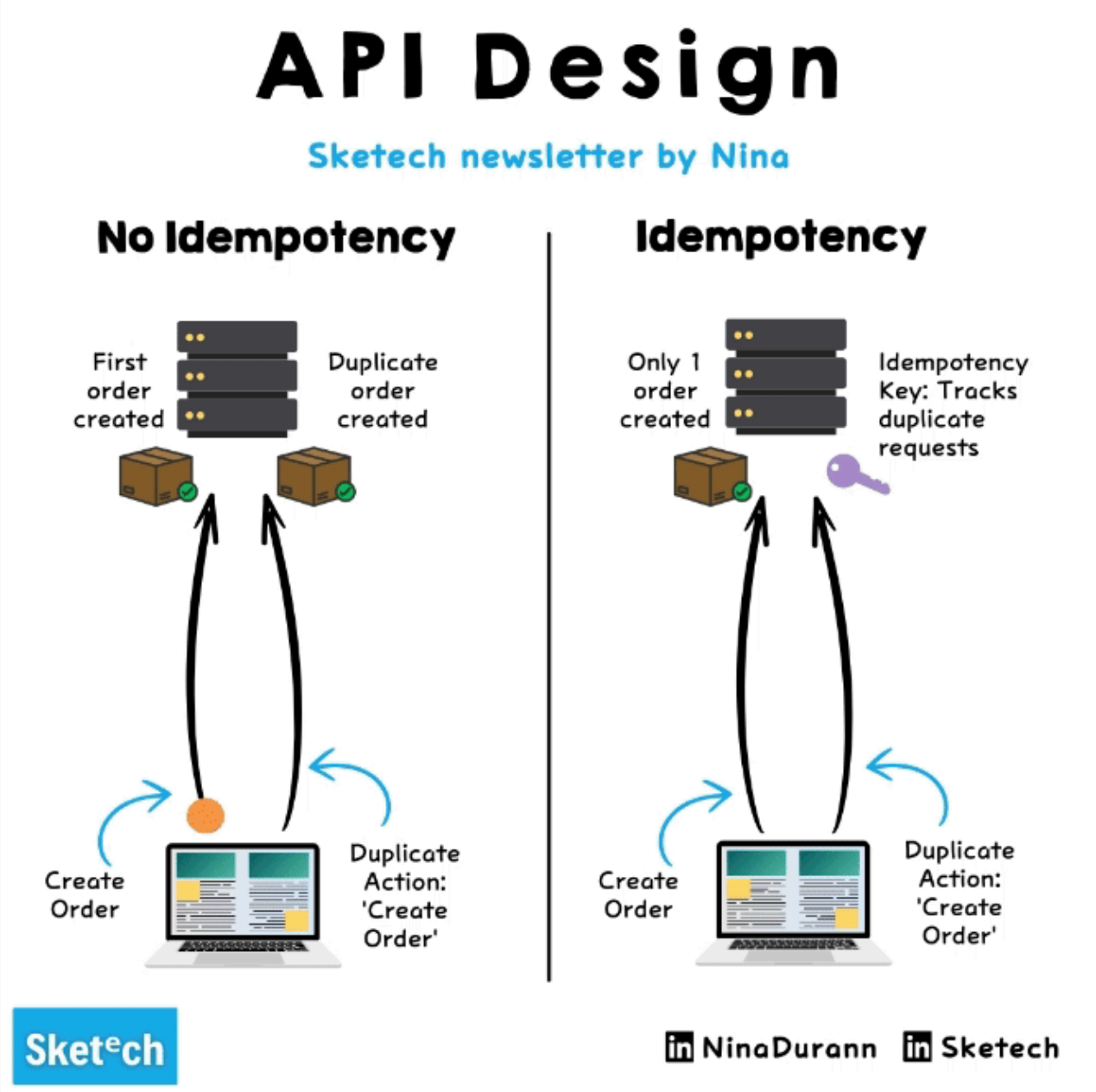
```rust
#[tokio::test]
async fn you_must_be_logged_in_to_access_the_admin_dashboard() {
    // Arrange
    let app: TestApp = spawn_app().await;

    // Act
    let response: Response = app.get_admin_dashboard().await;

    // Assert
    assert_is_redirect_to(&response, location: "/login");
}
```

```
NORMAL  trunk!  tests/api/login.rs  rust-analyzer      rust  utf-8[unix]  83% ln :49/59≡%:1
```

BDX I/O

D'ailleurs… Il faut qu'on parle de Wiremocks 🤩

Un cas difficile à tester : L'idempotence



Résultat, avec Wiremocks:

```rust
#[tokio::test]
async fn newsletter_creation_is_idempotent() {
    // Arrange
    let app: TestApp = spawn_app().await;
    create_confirmed_subscriber(&app).await;

    Mock::given(matcher: path("/email")) MockBuilder
        .and(matcher: method("POST")) MockBuilder
        .respond_with(responder: ResponseTemplate::new(200)) Mock
        // Assert made at Drop
        .expect(1) Mock
        .mount(&app.email_server) impl Future<Output = ()>
        .await;

    // Act - Part 1 - Login
    app.test_user.login(&app).await;

    // Act - Part 2 - Post new issue
+-- 16 lines: let id = Uuid::new_v4().to_string();·····

    // Act - Part 3 - Post duplicate request
+--  9 lines: assert_is_redirect_to(·····

    // Assert made at Drop (See Mock)
} async fn newsletter_creation_is_idempotent
```

BDX I/O

Ça, c'est le moment où j'ai décidé de me carciniser 🦀

Un écosystème complet

Il faut qu'on parle de sqlx

```
#[tracing::instrument(name = "Retrieve confirmed subscribers from the database", skip(pool))]
pub async fn get_confirmed_subscribers(
    pool: &PgPool,
) → Result<Vec<Result<ConfirmedSubscriber, anyhow::Error>>, anyhow::Error> {
    let confirmed_subscribers = sqlx::query!(
        r#"
        SELECT email from subscriptions WHERE statsu = 'confirmed';
        "#,
    type annotations needed
    cannot infer type (rustc E0282)
    ------------------------------------------------------------------
    https://doc.rust-lang.org/error-index.html#E0282
    ------------------------------------------------------------------
    error returned from database: column "statsu" does not exist

    Related information:

      * mod.rs#328,9: Actual error occurred here

     (rustc)
}
```

BDX I/O

Et ça peut même être très complexe !

```rust
#[tracing::instrument(name = "Retrieve presentations from the database", skip(pool))]
pub async fn get_presentations(
    pool: &PgPool,
    product_id: Uuid,
) -> Result<Vec<ProductPresentationSimple>, anyhow::Error> {
    let presentations: Vec<ProductPresentationSimple> = sqlx::query!(
        r#"
        SELECT Q.id, Q.unit_name, Q.unit_size, Q.packaging_name, Q.packaging_factor,
            Q.alternative_unit_name, Q.alternative_unit_factor, Q.created_at, Q.updat
        FROM products_presentations Q
        LEFT OUTER JOIN products_presentations_crosstable X
            ON X.presentation_id = Q.id
        LEFT OUTER JOIN products P
            ON P.id = X.product_id
        WHERE P.id = $1;
        "#,
        product_id
    )
    .fetch_all(pool)
    .await
    .unwrap()
    .into_iter()
    .map(|r| ProductPresentationSimple {
        id: r.id,
        unit_name: r.unit_name,
        unit_size: r.unit_size as u16,
        packaging_name: r.packaging_name,
        packaging_factor: r.packaging_factor as u16,
        alternative_unit_name: r.alternative_unit_name,
        alternative_unit_factor: r.alternative_unit_factor as u16,
        created_at: r.created_at,
        updated_at: r.updated_at,
    })
    .collect();

    Ok(presentations)
} async fn get_presentations
```

BDX I/O

Les autres poids lourds :

serde/deser
(sé/désérialisation)
tera (templating HTML)
chrono/jiff (le temps !)
et bien, bien d'autres...

Montons encore d'un cran...

Les Frameworks haut-niveau de Rust 🦀

BDX I/O

Axum

```rust
use axum::{ routing::{get, post}, http::StatusCode, Json, Router, };
use serde::{Deserialize, Serialize};

#[tokio::main]
async fn main() {
    tracing_subscriber::fmt::init(); // initialize tracing
    let app = Router::new()
        .route("/users", post(create_user)); // `POST /users` goes to `create_user`
    let listener = tokio::net::TcpListener::bind("0.0.0.0:3000").await.unwrap();
    axum::serve(listener, app).await;
}

#[derive(Deserialize)]
struct CreateUser { username: String, }

#[derive(Serialize)]
struct User { id: u64, username: String, }

async fn create_user(
    Json(payload): Json<CreateUser>, // Parse the request body as JSON into a `CreateUser` type
) -> (StatusCode, Json<User>) {
    let user = User { id: 1337, username: payload.username, };
    (StatusCode::CREATED, Json(user)) // Converts into a JSON response with 201 HTTP code
}
```

Actix-web

JSON

`Json<T>` allows deserialization of a request body into a struct. To extract typed information from a request's body, the type `T` must implement `serde::Deserialize`.

```
use actix_web::{post, web, App, HttpServer, Result};
use serde::Deserialize;

#[derive(Deserialize)]
struct Info {
   username: String,
}

/// deserialize `Info` from request's body
#[post("/submit")]
async fn submit(info: web::Json<Info>) -> Result<String> {
   Ok(format!("Welcome {}!", info.username))
}
```

BDX I/O

Et le monde des Rustacé-es(c) va vite, très vite… 🏃💨

Dioxus, le framework isomorphique !

Introduction

Dioxus is a framework for building cross-platform apps with the Rust programming language. With one codebase, you can build apps that run on web, desktop, and mobile platforms.

src/readme.rs

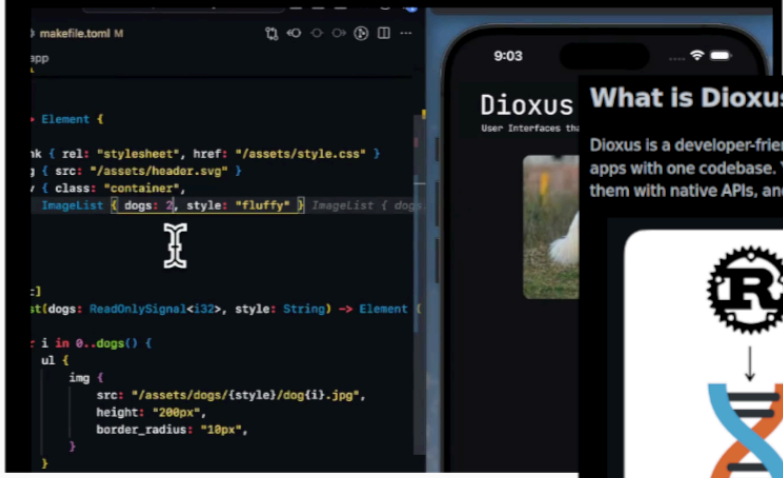
```rust
use dioxus::prelude::*;

pub fn App() -> Element {
    let mut count = use_signal(|| 0);

    rsx! {
        h1 { "High-Five counter: {count}" }
        button { onclick: move |_| count += 1, "Up high!" }
        button { onclick: move |_| count -= 1, "Down low!" }
    }
}
```

BDX I/O

On peut littéralement tout faire avec 🤯

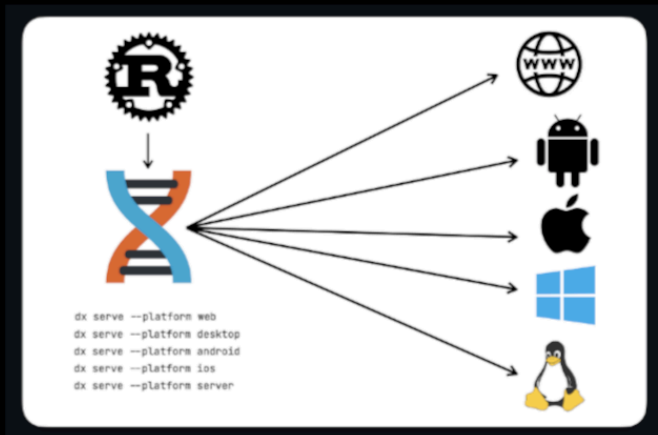
Stellar Developer Experience

With Dioxus, we try to maintain a very high bar for developer experience. We believe that building apps should be fun and straightforward. We've worked to push forward Rust itself, developing technologies like Subsecond Rust hot-reloading, WASM bundle-splitting, linker-based asset bundling, and a modular WGPU-based HTML/CSS renderer.



What is Dioxus?

Dioxus is a developer-friendly framework that empowers developers to ship cross-platform apps with one codebase. You write your apps in Rust, style them with HTML/CSS, enhance them with native APIs, and distribute them as platform-native bundles.

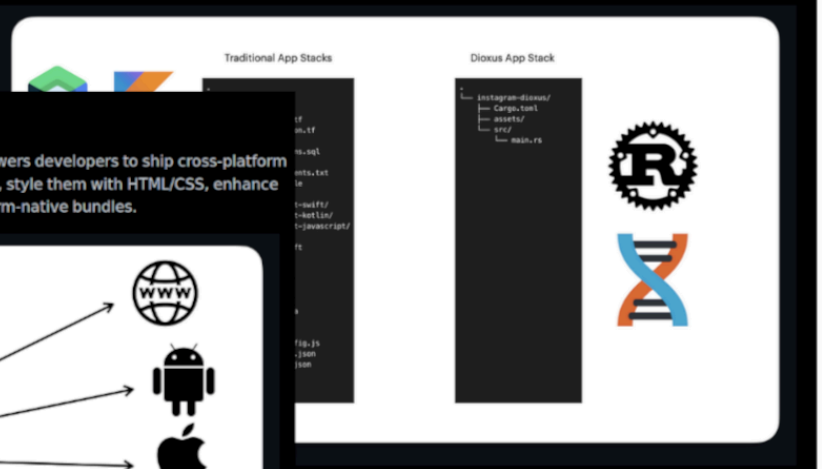


In many ways, Dioxus is similar to Flutter: we integrate our own build tools, foster an ecosystem, and provide a GUI framework. In key areas, Dioxus takes a different approach:

- Apps are declared with HTML and CSS instead of a custom styling solution
- Reactivity is inspired by web frameworks like React and SolidJS
- Dioxus code runs natively with no virtual machine, enabling zero-overhead calls to system APIs

Why Dioxus?

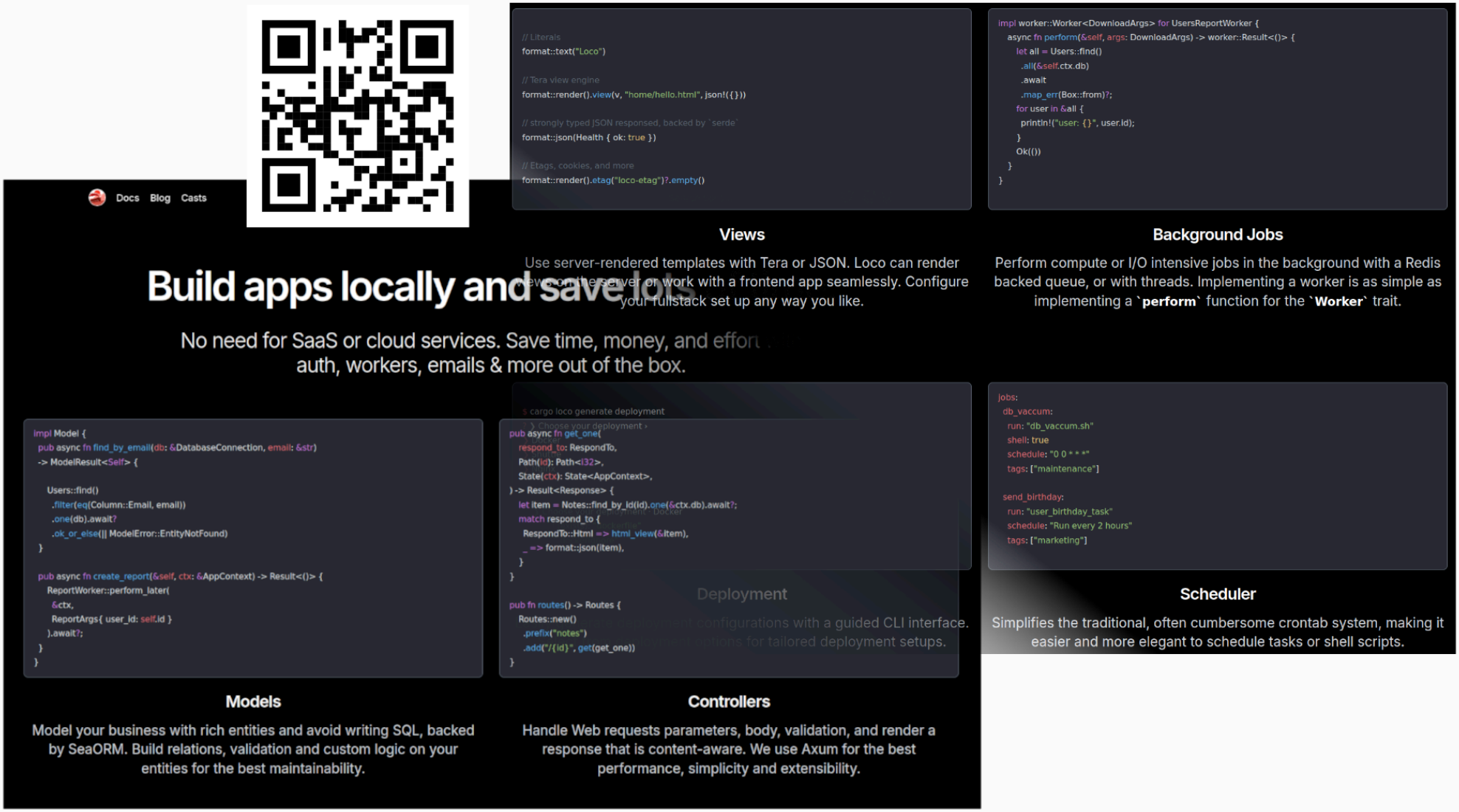
We built Dioxus because we believe the current standard of building apps is too complex. Developers need to learn and install dozens of different tools just to get their app into the world. We need a simpler and more powerful framework to bring apps from ideas to reality.



k that is fast, flexible, and has a minimal learning curve.
launch their app from idea to production as fast as
ls and a simpler architecture make it easier to develop
ship faster and are more likely to succeed.

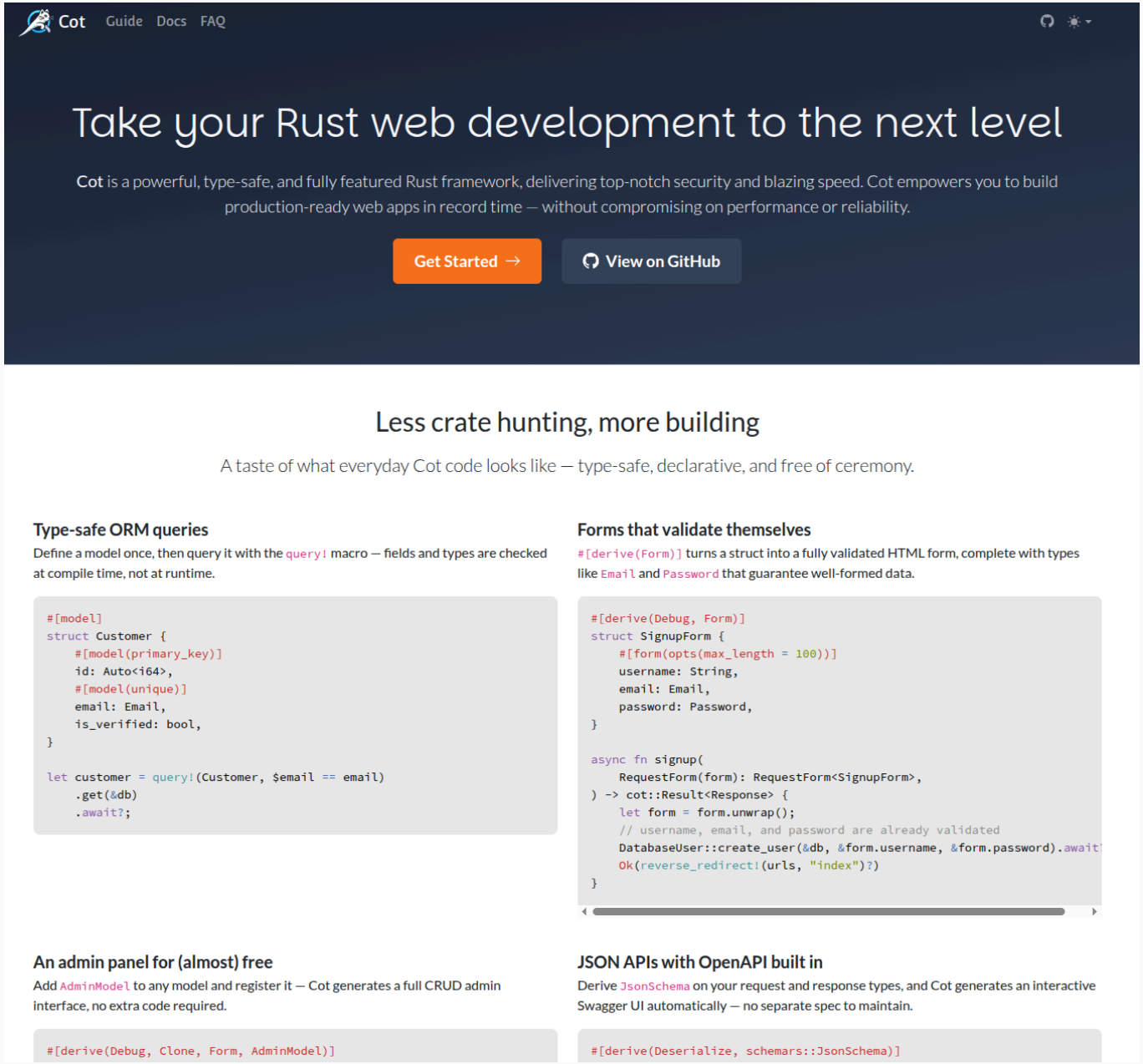
BDX I/O

Vous venez plutôt de Rails et RoR ? No problemo…y a Loco 🚂



BDX I/O

Il y en a pour tous les goûts ! 🐔

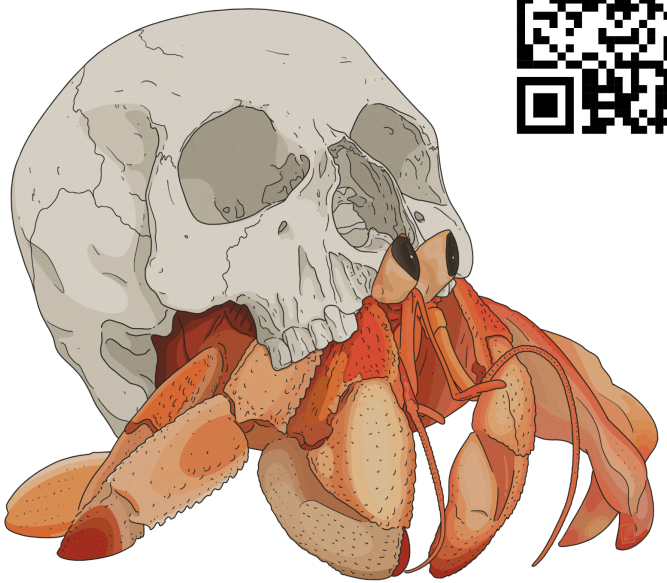


Si vous voulez aller plus loin...

(parce qu'il y en a encore plein, des dingueries comme ça 😅, dans Rust)

ZERO TO PRODUCTION IN RUST

AN OPINIONATED INTRODUCTION TO BACKEND DEVELOPMENT



LUCA PALMIERI

Learn Rust With Entirely Too Many Linked Lists

I fairly frequently get asked how to implement a linked list in Rust. The answer honestly depends on what your requirements are, and it's obviously not super easy to answer the question on the spot. As such I've decided to write this book to comprehensively answer the question once and for all.

In this series I will teach you basic and advanced Rust programming entirely by having you implement 6 linked lists. In doing so, you should learn:

- The following pointer types: `&`, `&mut`, `Box`, `Rc`, `Arc`, `*const`, `*mut`, `NonNull`(?)
- Ownership, borrowing, inherited mutability, interior mutability, Copy
- All The Keywords: struct, enum, fn, pub, impl, use, ...
- Pattern matching, generics, destructors
- Testing, installing new toolchains, using `miri`
- Unsafe Rust: raw pointers, aliasing, stacked borrows, UnsafeCell, variance

Yes, linked lists are so truly awful that you deal with all of these concepts in making them real.

Everything's in the sidebar (may be collapsed on mobile), but for quick reference, here's what we're going to be making:

1. A Bad Singly-Linked Stack
2. An Ok Singly-Linked Stack
3. A Persistent Singly-Linked Stack
4. A Bad But Safe Doubly-Linked Deque
5. An Unsafe Singly-Linked Queue
6. TODO: An Ok Unsafe Doubly-Linked Deque
7. Bonus: A Bunch of Silly Lists

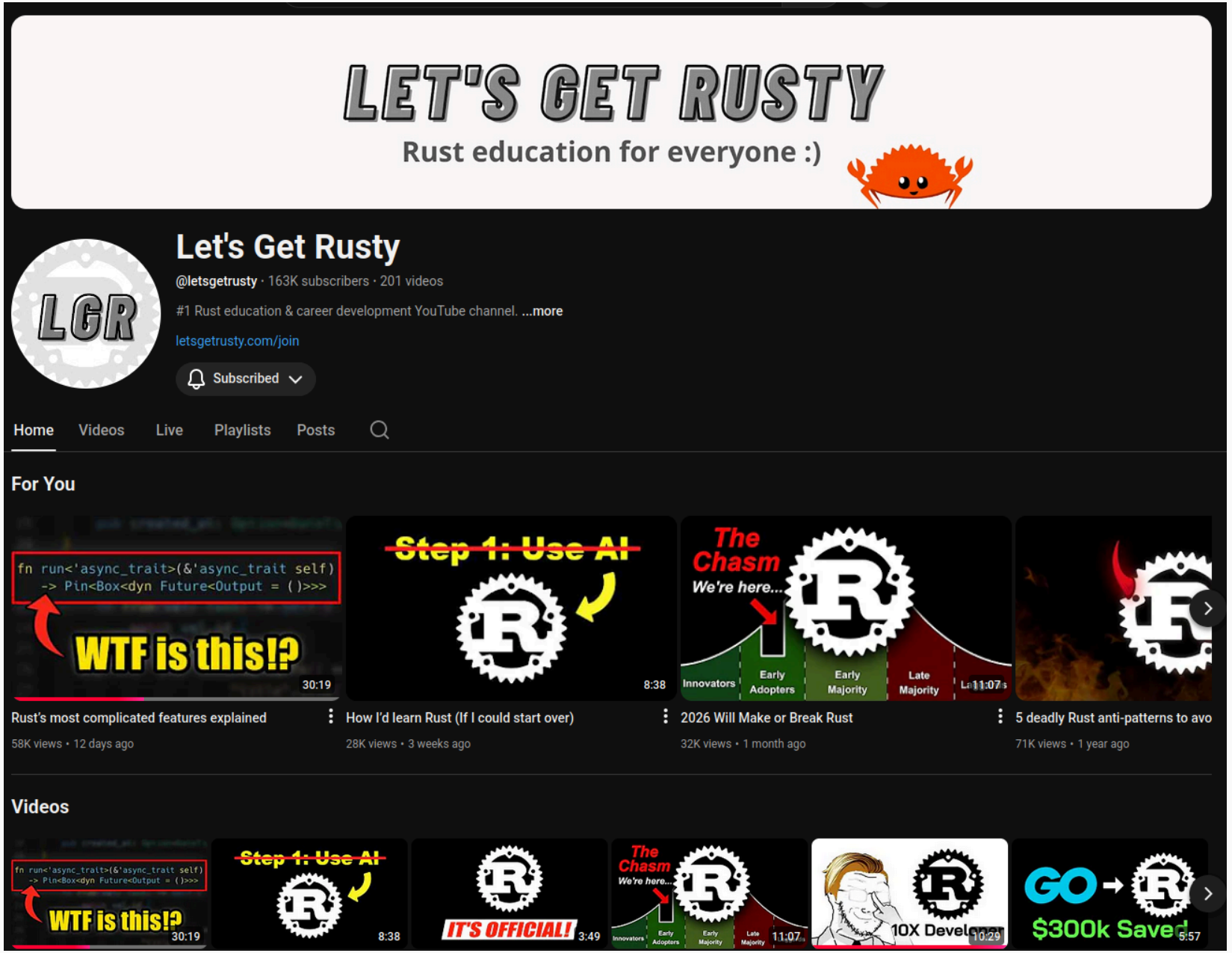
Benjamin Geer

Créer une application hypermédia en Rust

Aperçu de l'application

L'application qui sert d'exemple dans Hypermedia Systems est une application CRUD qui gère une liste de contacts. Le livre montrer comment créer une application hypermédia qui offre bonne interface utilisateur, en alternative aux applications monopages. Grâce à une minuscule bibliothèque JavaScript (htmx) et à quelques attributs déclaratifs sur les éléments HTML, toutes sortes d'interactions utilisateur peuvent récupérer des fragments de code HTML depuis le serveur et ne modifier que certaines parties de la page. Par exemple, quand on passe d'une page à une autre dans la liste des contacts ou qu'on supprime un contact, la liste affichée s'actualise sans que la page ne soit rechargée.

YouTube: Let's Get Rusty !



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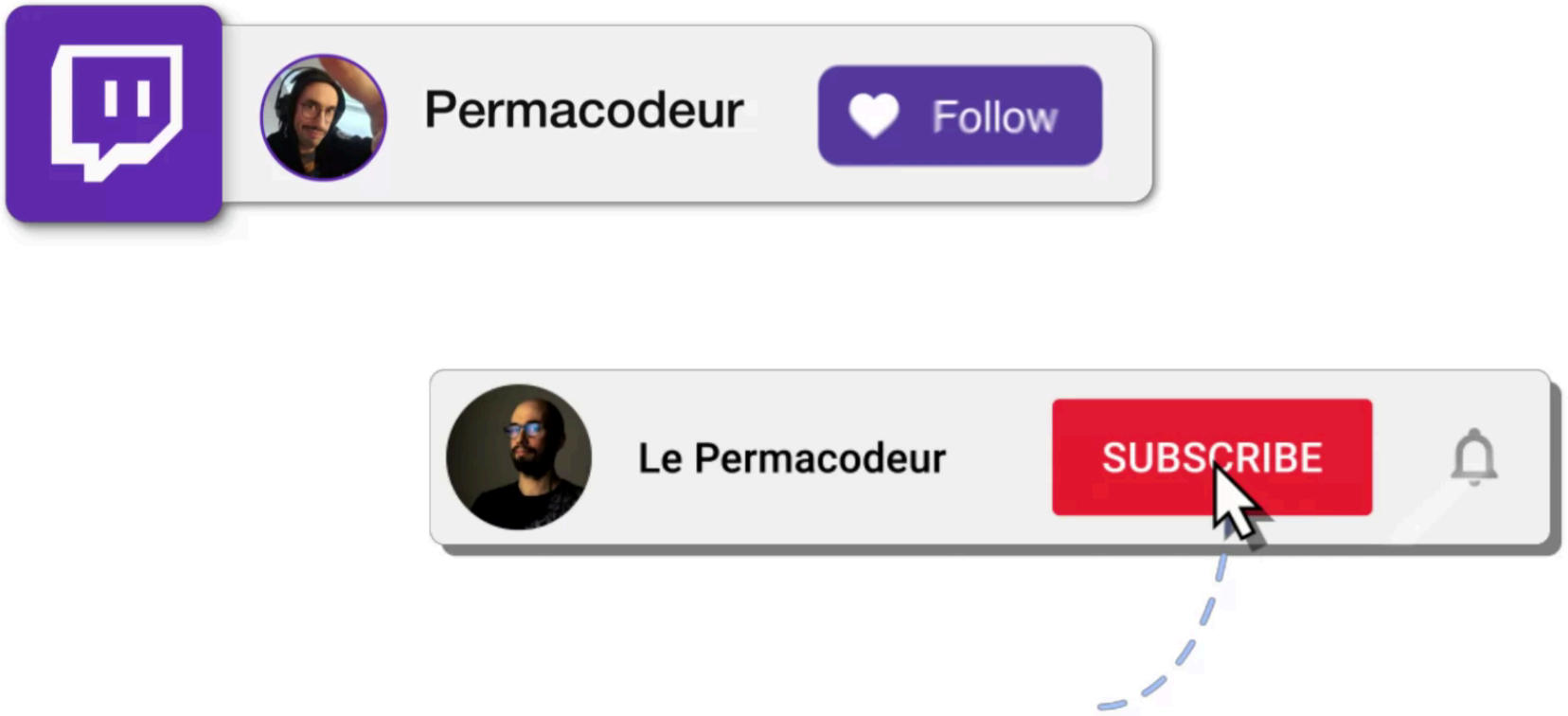
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🤩 Advent Of Code 2025...in Rust ! Jour 8/12 - Partie 1 🦀

```
fold(junction_box_list: &Vec<Coords>, connection_count: usize) -> usize {
    let mut nearest_list: BTreeMap<Coords, Coords> = BTreeMap::new();
    let mut nearest_list_sorted: HashMap<(Coords, Coords), i64> = HashMap::new();
    let max_distance: i64 = i64::MAX;

    for junction_box: &(u32, u32, u32) in junction_box_list {
        for other_box: &(u32, u32, u32) in junction_box_list {
            if junction_box < other_box {
                nearest_list_sorted.insert(
                    k: (*junction_box, *other_box),
                    v: compute_euclidian_distance(first_box: junction_box, second_box: other_box),
                );
            } else if junction_box > other_box {
                nearest_list_sorted.insert(
                    k: (*other_box, *junction_box),
                    v: compute_euclidian_distance(first_box: junction_box, second_box: other_box),
                );
            }
        }
    }
    let nearest_list_sorted = ne

    println!("Nearest list: {
    println!();
    println!("Nearest list le

    println!("Nearest list, s
    for (couple: &((u32, u32,
        println!("{distance}
    }
```

```
nearest_list_sorted HashMap<((u32, u32, u32), (u32, u32, u32)), i64, RandomState> v [LS]
nearest_list BTreeMap<(u32, u32, u32), (u32, u32, u32), Global>                    v [LS]
needs_drop()~(use std::intrinsics::needs_drop) const fn() -> bool                  f [LS]
needs_drop()~(use std::mem::needs_drop) const fn() -> bool                         f [LS]
net(use core::net)                                                                 M [LS]
net(use std::net)                                                                  M [LS]
net(use std::os::linux::net)                                                       M [LS]
net(use std::os::unix::net)                                                        M [LS]
new                                                                                [A]
```

```
INSERT  trunk!  src/main.rs[+]  rust-analyzer                rust  utf-8[unix]  36% ln:123/334≡%:33  W:2(L104)
-- INSERT --
```

```
test result: FAILED. 0 passed; 1 failed; 0 ignored; 0 measured; 3 filtered out; finished in 0.00s

stderr ———

thread 'tests_fold::fold_sample' (1550824) panicked at src/main.rs:310:9:
assertion `left == right` failed
  left: 28
 right: 40
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace

Cancelling due to test failure: 1 test still running
Cancelling [ 00:00:06] [                                        ]  3/4: 1 running, 0 passed, 0 skipped
           [ 00:00:06] aoc2025::bin/aoc2025 tests_fold::fold_gouz
```

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distances, je les trie, et pour les 1000 premières : C'est la fonction de connexion la clé où faut être sur de faire tous les cas : a et b n'ont pas de circuit, a n'a pas de circuit, b oui, b n'a pas de circuit, a oui, a et b ont un circuit (on merge toutes les boxes du circuit de b dans le circuit de a)
Alexandre_grdl: Attends, la partie 1 c'est de faire 1000 connexions,
Alexandre_grdl: Tu fais 1000 connexions ou tu traites 1000 boxes ?
Brack_0: @Alexandre_grdl yep pareil
Brack_0: j'ai juste un algo dédié pour la tambouille que tu décris Alexandre
Alexandre_grdl: Je comprends pas trop la boucle que tu fais. Parce qu'en fait, très bêtement, si je veux faire tous les couples possibles sans doublons , tu fais for i = 0 ; i < taille; i++ { for j = i+1 (très important); j < taille; j++)
Alexandre_grdl: là tu vas avoir les doublons. (a,b) et (b,a)

t _ n e a r ↓ ⏎ _ = _ n e

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Mes liens 🤓

> *Rust 🦀 a désormais tout ce qu'il faut pour délivrer de la valeur pour vos applications métier. Jetez-y un œil ! 👀*